

Codebase Assessment & Spark Appliance Release Plan

A management-level map of what the software is, how healthy the code is, and a costed plan to turn the current build into a first releasable, hardware-locked appliance for the DGX Spark. **Version 2.0 rebuilds the July 24 assessment on measured evidence** — a live end-to-end benchmark, twelve findings from hands-on operator testing, and the arrival of the β 0.8.5 release candidate.

Prepared for Vitaly Bondarenko and project stakeholders · 27 July 2026 · Sources: seven-part read-only source review (24 July, commit `beca6ff`), Wave-1 end-to-end benchmark (25 July), manual QA findings MF-1...12, and the β 0.8.5 release notes (27 July).

53%	0	31	3
of staged fights raised an alert — measured at shipped default settings, not estimated	false positives across quiet and busy background channels over a two-hour run	open findings — 12 manual, 6 from the automated bench, 13 from code review	database migrations in 0.8.5 — the first schema change since the pilot shipped

Executive summary

EVA AI is an AI layer over the Luxriot Evo video system: it describes every camera scene with a vision model, raises alerts, makes footage searchable, and answers operator questions with evidence. **The July assessment holds.** The runtime engineering is strong — durable data handling, database-enforced tenant isolation, a read-mostly agent with human-approved writes, a dense regression suite for the core, and mature install and rollback tooling. It runs on the Spark.

What version 2.0 adds is **evidence where there was inference**. The system has now been measured end to end against staged footage with known ground truth, driven by hand for several days by an operator, and handed a new release candidate. Three things came out of that, and they reshape the plan:

One — detection at default settings is a policy problem, not a perception problem. The vision model missed roughly half the staged fights, but produced zero false alarms on quiet and busy background scenes and never mistook dancing for violence. The model sees correctly; the thresholds and severity policy that decide what becomes an alert are miscalibrated in both directions — too quiet on real events, uniformly loud on trivial ones.

Two — the "silent evidence loss" risk is not theoretical and not a 50-channel problem. July 24 flagged it as something that would appear at target scale. The benchmark shows the drop counters already non-zero on *every one* of nine channels. Whether those are benign coalescing or true evidence gaps is the single most important technical question in this document.

Three — 0.8.5 breaks the "code-only upgrade" assumption the appliance update path was built on. Every release since the pilot has been a code overlay against an unchanged schema. 0.8.5 requires three database migrations, forbids the previous field-upgrade recipe, and its generic updater deliberately refuses to migrate. The appliance update story now has to carry a database with it.

What this document is — and is not

A synthesis of a static source review at commit `beca6ff`, a two-hour instrumented benchmark at nine channels, several days of hands-on operator testing, and the 0.8.5 release notes as written. It does *not* reflect a soak at the 50-channel target, an installed and accepted 0.8.5, or an independent audit of the 0.8.5 claims — those close only against the deployed appliance and are called out where they matter.

From reasoned risk to measured fact

Version 1.0 was a static read of the source at one commit. It was careful to say what it could not know. Three days of measurement and one release candidate later, most of those unknowns have an answer — and the answers are mostly, but not entirely, reassuring.

WHAT V1.0 SAID (24 JULY, FROM SOURCE ALONE)	WHAT WE NOW KNOW (27 JULY, MEASURED)	EFFECT ON THE PLAN
Upgrades are data-compatible; the schema has not moved since the pilot	Superseded. 0.8.5 requires three Alembic revisions to head <code>20260726_0009</code> , explicitly forbids the 0.8.4 field-upgrade script as a recipe, and ships a generic updater that refuses to migrate and gates on seeing the new head.	Point ② (update path) needs a database story. Material.
Under load, a timed-out vision batch is dropped with no marker — expected at 50 channels	Confirmed early. At only nine channels over 2.07 h, every channel logged dropped batches (47 in total) and dropped frames (564). The mechanism the review predicted is already running.	Promoted to the top of the "should fix" tier; needs Sasha's read. Material.
Model-output quality is out of scope and unassessed	Now measured. 53% recall on staged fights, 0% false-alarm on dancing, 0 false positives on quiet and background scenes, 0.198 s timestamp accuracy, 0.5 s end-to-end latency.	New page 5. Changes what "ready" means. Material.
The 16.5k-line browser UI has zero automated tests	Unchanged, and now quantified. A few days of hands-on use produced 12 findings, four of them major; five sit in the login form and settings dialog alone.	Confirms the UI-test workstream's priority.
~110 threads in one process at 50 channels is unproven	Still unproven. Nine channels at ~110 summaries per channel-hour is the only measured point on the curve. 0.8.5 explicitly declines to claim its own 4–6 stream target in advance.	Soak remains the gate. Unchanged.
Phase-6 acceptance on the Spark is the open gate	Still open — and now load-bearing. 0.8.5's migration procedure requires the box to be a verified 0.8.4; the Spark runs branch HEAD, thirteen commits past the reviewed release candidate.	Now blocks the 0.8.5 upgrade too. Escalated.
Four legal and honesty blockers must close before release	All four still open. No licence file, no written model acceptable-use confirmation, auth still off by default in code, audit hash-chain and tool ledger still schema-only.	Unchanged, and unaffected by 0.8.5.

What the evidence vindicated

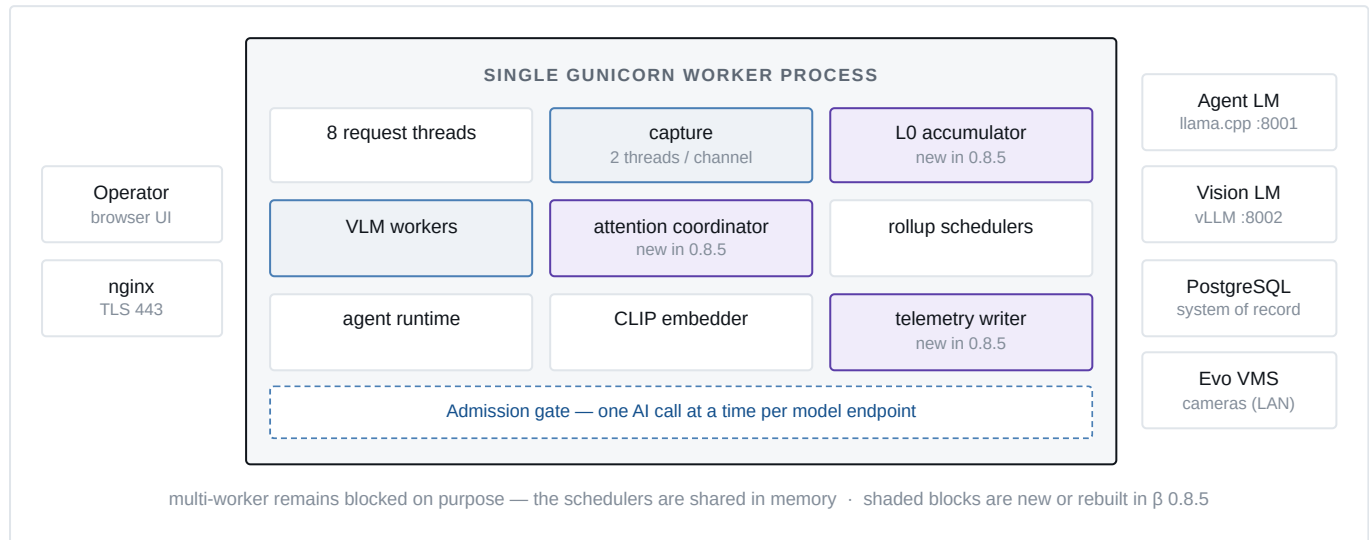
The review's read of the runtime was accurate. Tenant isolation, the admission gate, the read-mostly agent and the durability design all behaved as described. Nothing in three days of adversarial poking contradicted the architecture chapter — the failures found were in *defaults, policy and the front end*, exactly where the review said the thin ice was.

What it caught the review could not

Two features fighting each other (adding probes to a channel silently lowered that channel's alert sensitivity), and a saved setting that never reaches a running capture session. Both are invisible in source and only appear when the system is driven. Both are the argument for the automated bench existing at all.

One product, one process — and a rebuilt attention layer

The shape is unchanged: a Python application whose web UI, capture threads, vision workers, schedulers and agent all run as threads inside a single process, surrounded by a database, a TLS front door and two local model servers. What 0.8.5 changes is everything about *how frames become evidence*.



How a frame became evidence in 0.8.4

A snapshot every ~5 seconds per channel. A per-second filter kept the single most useful frame. Twelve of those formed a batch sent to the vision model. Its written summary and any alerts landed in one database table, and rolled up into 15-minute, hourly and 6-hour narratives.

How it works in 0.8.5

Dense computer vision still runs at the configured stream rate, but those raw frames are no longer archived. Each one-second bucket collapses to a compact motion interval. Admission becomes **activity-sensitive**: one frame every 5 s when quiet, every 2 s in normal activity, every 1 s in a burst — capped at 16 snapshots per request and flushed after at most 60 seconds. Rollups become 15-minute, one-hour and **eight-hour** layers.

Why the change matters commercially

The old pipeline spent the same vision budget on an empty corridor as on a fight. The new one spends it where motion is, and — importantly for a fixed-hardware appliance — makes the ceiling *declarable*: a hard 16-snapshot cap and a hard 60-second flush are numbers you can put in a datasheet. This is the single most appliance-friendly change in the release.

Three things that still hold

It is one process. A crash takes everything down together; it scales up, not out. **One queue governs throughput** — every AI call passes an admission gate running one call at a time per model, and the vision model remains the real capacity budget. **The agent is read-mostly** — it searches everything but changes only six things, each behind a human preview-approve-apply step.

Watch item. The rollup depth moved from 6 to 8 hours and a new durable telemetry writer shares the process. Both add background work to the same GPU and the same single process that the benchmark already observed slowing down over hours.

Strong runtime, thin release layer — now with two moving parts

Twelve dimensions from the July review, re-rated against everything learned since, plus one new dimension that only became rateable once the system was measured. The pattern is unchanged: everything about *running* the software rates well; everything about *shipping and sustaining* it is where the work is.

● Green — a real strength, ship as-is ● Yellow — usable, needs attention ● Red — must be built or fixed ▲ / ▼ movement since 24 July

DIMENSION	RATING	Δ	WHAT THE EVIDENCE NOW SAYS
Data durability	● Green	—	Rows before caches, atomic state files, rollups rebuild from surviving data. Design unchanged and uncontradicted by the run.
Security model	● Green	—	Database-enforced tenant isolation and a default-deny agent gateway both held. Zero 403-class surprises in the benchmark.
Agent safety	● Green	▲	Read-mostly with human-approved writes. 0.8.5 adds a hard 10-call ceiling on video research and duplicate-call suppression — bounded by construction, not by prompt.
Test coverage — core	● Green	—	0.8.5 reports 782 passed / 23 skipped. Dense and growing — but the skip count moved from 18 to 23 with no stated reason, which is exactly the drift the per-release check exists to catch.
Operations tooling	● Green	—	~7,600 lines of install, preflight, rollback and verify scripting with offline delivery and a rehearsed rollback.
Detection quality new dimension	● Yellow	NEW	Perception good, policy poor. Zero false positives and zero dance-as-violence errors, but only 53% of staged fights alerted, and everything that did alert came out at one severity. Fixable in thresholds and policy, not in the model.
Offline / air-gap posture	● Yellow	—	Zero telemetry and no runtime internet calls — still leaking through a Google Fonts import and unguarded model loaders. Both remain half-day fixes.
Config & defaults	● Red	▼	Downgraded. Shipped probe thresholds match everything on every channel; auth is off by default; and a saved setting does not reach a running capture session — it silently caches at start. Defaults are where this product currently fails its user.
Architecture / modularity	● Yellow	—	Clean structure with no tangles — but four 12–17k-line files still concentrate the whole system, and 0.8.5 adds substantial new subsystems into the same process.
Docs accuracy	● Yellow	—	Curated docs with a drift check — but the AI orientation file still describes the predecessor product and 60 settings are undocumented. 0.8.5 adds a new configurable surface to document.
Dependency & licence hygiene	● Yellow	—	Permissive, boring stack — but still no licence file, one unknown-licence package, and no version lockfile.
Test coverage — UI	● Red	—	16,500 lines, zero automated tests, no framework. Twelve manual findings in days of use is the cost of that, measured.
Continuous integration	● Red	—	None exists. With 0.8.5 introducing migrations, the absence of an automated upgrade-and-rollback test is now a sharper risk than it was a week ago.
Bus factor	● Red	—	One author; deep rationale lives in the author's notes and remotes rather than in CI, comments or an accurate orientation file.

The through-line, restated for v2.0. You are still not being asked to fix a shaky product — you are being asked to wrap a solid runtime in the packaging, protection and guard-rails an appliance needs. What changed is that *defaults and policy* have joined *continuity* as the second front. A sealed appliance ships its defaults to every customer permanently; on current evidence those defaults would give a first-time user a firehose of identical high-severity alerts about bicycles and silence about half the fights.

The first numbers this product has ever had

A two-hour instrumented run, nine channels of staged footage with frame-exact ground truth, at shipped default settings. This is the out-of-the-box customer experience, measured — not a tuned demo.

10_{/19}

staged fights that raised an alert —
53% recall

0_{/17}

dance cycles misread as violence —
no confusion of energetic motion
with fighting

0

false positives on quiet and busy
background channels

50_s

median alert latency (7 s best, 97 s
worst)

Reading the recall number correctly

53% is the honest headline, but the shape of the errors matters more than the rate: the system produced **no false alarms at all** — not on an empty corridor, not on a construction crane, not on dancing a naive motion detector would flag instantly. A model that hallucinated violence would have failed differently. That asymmetry says the vision model perceives the scenes correctly and the *decision layer* is set too conservatively — a threshold-and-prompt problem measured in days, not a model problem measured in months.

FIGHT CHANNEL · ALERT PER STAGED CYCLE

Alerted (10 cycles)

53%

Missed (9 cycles)

47%

A follow-up experiment with probes disabled reached 5 of 7 — the best result recorded — which is itself evidence for the probe-interference finding opposite.

The severity flood

Every single alert the bicycle channel produced — 18 cycles, 21 alerts — came out at **HIGH** severity. A separate measurement put the sustained rate at **9.3 alerts per hour from one cyclist**. Meanwhile the actual fights, when they alerted, arrived at the same severity as the cyclist.

An operator watching this product at defaults would learn within one shift that HIGH means nothing. That is the classic path to a monitoring system being ignored, and it is a policy change, not a code change.

Dropped batches at nine channels

Over 2.07 hours the nine channels logged **47 dropped batches and 564 dropped frames** — every channel non-zero. The July review documented exactly this path: with the durable inference queue off (the default), an admission timeout destroys a vision batch and leaves no coverage-gap marker.

The open question for Sasha: are these benign coalescing events, or are they silent holes in the evidence record? On a product sold on evidence, the answer decides whether the durable queue must ship on by default.

Timing is excellent

Median timestamp error **0.198 s** against optical ground truth and **0.5 s** end-to-end latency. Evidence lands where it belongs on the timeline — the foundation any forensic claim rests on.

Coverage is even

Roughly **110 summaries per channel-hour**, clustered from 102 to 116 across all nine channels. The budget is distributed predictably — which is what makes a published channel ceiling achievable.

Two features fighting

Creating probes on a channel **stopped its alerts within a minute**; disabling them restored alerting within two. Adding a probe unknowingly lowers that channel's sensitivity, with no warning.

What this page buys you. Three days ago the honest answer to "how well does it work?" was "the code looks sound". It is now a number with a method behind it, repeatable in about two hours, that will move measurably when the thresholds change — the difference between a demo and a product.

Four true blockers, then the calibration work

Ranked by whether they block a public release, how likely they are to bite, and how much effort they take. The first tier is unchanged from July and unaffected by 0.8.5. The second tier has been reordered by evidence — two items moved up because they were measured rather than predicted.

MUST FIX BEFORE RELEASE

RISK	IMPACT	EFFORT
No licence / notice file	Without a declared licence, recipients have no legal grant to use the software; dependency notices are undisclosed. Blocks distribution outright.	1–2 days
Model acceptable-use (Qwen)	The model weights — not the code — are the real legal exposure for government camera use. Needs written confirmation that the specific checkpoints permit it.	Legal, elapsed
Insecure-by-default settings	Authentication is off by default; with it off the agent gateway is bypassed entirely. The appliance image must make secure the default rather than the exception.	2–3 days
Compliance-claim honesty	The tamper-evident hash-chained audit log and the per-call tool ledger exist as empty database columns only. Do not claim them until built, or correct the wording.	Hours (doc) 1–2 wk (build)

SHOULD FIX BEFORE BROAD ROLLOUT — REORDERED BY MEASURED EVIDENCE

RISK	IMPACT	EFFORT
Alert policy & severity calibration UP	Measured, not predicted. Half of real events silent; trivial events at maximum severity 9× per hour; one severity for everything. This is the finding a customer meets on day one. Threshold, prompt and severity-mapping work — plus a way to tune it per site.	3–5 days + re-baseline
Silent evidence loss UP	Observed at nine channels, not fifty. 47 dropped batches in two hours with no coverage-gap marker. Turn the durable inference queue on in production and emit a gap row when a batch dies.	1–2 days + validation
Settings that do not apply	A running capture session caches configuration at start; a setting saved in the UI reads back as on and does nothing. It silently disabled the product's only push-to-operator alerting path for an entire 45-minute measurement window.	2–3 days
Probe / alert interference	Enabling probes on a channel suppressed that channel's alerting on marginal events; disabling them restored it. Two shipped features degrade each other with no warning. Mechanism unconfirmed — needs one replication and an engineering read.	Investigation 2–4 days
Unproven at target channel count	~110+ threads in one process at 50 channels, still never load-tested. 0.8.5 declines to claim even its 4–6 stream target in advance. Run the soak and record the ceiling.	2–3 days
Retention & disk handling	Cleanup runs only while writing; a cap path leaks image files; a full disk is an unhandled crash. On a sealed appliance with no shell access, a full disk is a truck roll.	2–4 days
No CI / untested UI	No automated safety net. With migrations now in play, an automated upgrade-and-rollback test matters more than it did in July.	~1 week
Service credentials in plain text	Four secret-bearing values sit in the environment file, including the video-system service account — which currently holds administrator rights on Evo. Standard practice, weak posture for a security product under review.	1 day (perms) 2–3 d (sealed)
Cross-tenant skills injection	If multi-tenant, a shared agent-instruction folder lets one operator seed standing instructions into other tenants' sessions. Scope and audit it.	3–5 days*

Worth restating. The software still phones home nowhere, carries no telemetry, and needs exactly one write to the Evo system. That remains a certifiable baseline for an air-gapped, government-grade appliance once the paperwork and two small leaks are closed.

* if multi-tenancy is a real requirement — still an open stakeholder question.
Track opportunistically: self-host the three Google fonts and set the offline flags to make air-gap a code guarantee (~½ day); rewrite the stale AI orientation file (~½ day); normalise the author's three git identities; fix the eight smaller UI findings in the register.

A good release that quietly rewrites the update plan

β 0.8.5 is a substantial, well-documented release: bounded per-channel vision batching, durable attention telemetry, an explicit memory hierarchy, operator false-positive feedback with export, and a hard ceiling on agent tool loops. It is also the first release since the pilot that touches the database — and that single fact lands squarely on the appliance strategy.

What it gives the appliance plan

Declarable limits. A hard 16-snapshot cap and 60-second flush per channel turn "it depends" into datasheet numbers — exactly what point ⑤ needs to publish a channel ceiling.

The guard-rail monitor now has inputs. The readiness endpoint exposes attention storage state, queue, drop and failure counters, coordinator status and probe counts. In July the monitor would have had to invent its telemetry; now it consumes it.

Bounded agent work. A 10-call research ceiling and duplicate-call suppression make the agent's cost per question bounded by construction — a real constraint on a fixed-memory box.

Machinery for the calibration work. Operator false-positive annotation with five reasons, provenance and export gives the severity problem a data source it did not have. It is the instrument, not the fix — the severity policy itself is untouched.

What it costs the appliance plan

The update path must now carry a database. Three revisions to head 20260726_0009. The notes forbid the previous field-upgrade script and state that the generic bundle updater is *deliberately non-migrating* — it accepts the code overlay only once it already sees the new schema. On a sealed box with no shell, "run the migration by hand with a privileged connection string" is not a procedure that exists.

Rollback is now a database problem. Today's rehearsed one-command rollback restores code and configuration. A failed migration in the field needs a tested database rollback — or an update model that snapshots and restores the whole volume atomically.

The precondition does not match the Spark. The procedure requires a verified 0.8.4; the Spark runs branch HEAD, thirteen commits past the reviewed candidate, Phase-6 still open.

A new hardware target appeared. The known-limits section names a "4070-class host" and a "port deployment" as the capacity reference for 4–6 vision streams. Neither appears in the deployment profiles, and neither is the Spark.

QUESTION RAISED BY 0.8.5	WHY IT NEEDS AN ANSWER BEFORE THE APPLIANCE IMAGE IS DEFINED
Is the Spark still the appliance target?	A 4070-class host and a "port deployment" are now the stated capacity reference. If the first shipped box is not a Spark, points ①, ③ and ④ change platform underneath.
What is the appliance's migration model?	Updater gains migration rights · whole-volume A/B swap with rollback · migrations only at manufacture. Each implies a different amount of work in point ②.
Why did the skip count move from 18 to 23?	Five newly skipped tests with no stated reason, in the release that introduces migrations. The per-release check exists precisely to ask this.
Does the new attention storage close the dropped-batch gap?	Revision ...0008 stores scheduler decisions and attention episodes. If a dropped batch now leaves a durable trace, the top "should fix" item is partly closed by this release; if not, it stands.
Should alerts trip Evo external events?	0.8.5 confirms probes create no bookmarks and no notifications — alerting stays recording-notifying. One HTTP call to Evo's external-event endpoint would unlock its whole action palette. Still not adopted.

Net read on 0.8.5. The engineering is going in the right direction — bounded, declarative, telemetry-rich, and honest about what it has not proven. Nothing in it argues against the appliance strategy. But it converts point ② from "extend existing tooling" into "design a migration-safe update model", and that belongs in the estimate before anyone commits to a date.

The strategy still holds; two of five got harder

Your five points form one strategy: ship a sealed, hardware-bound box you control end to end. That remains a strong fit for this software. Below, each point is re-assessed against what three days of evidence and one release candidate changed.

① A first releasable version — Spark-only, appliance-only

Where it stands: the software installs and runs on the Spark, and Wave-1 proved the whole chain works end to end against real footage with sub-second timing accuracy. That is a stronger position than July: the box does not just start, it performs measurably.

What changed: two things. Phase-6 acceptance is still open and is now also the precondition for the 0.8.5 migration. And the release notes' capacity target names non-Spark hardware — so "Spark-only" needs an explicit confirmation before the image is built for it.

Also new: a sealed image ships its defaults permanently. On current measurement those defaults produce a severity flood and half-silent detection. **Calibration is now part of the image definition**, not a post-release tuning exercise.

Actions: confirm the hardware target; finish Phase-6; define the image build; flip secure defaults; bake model weights and offline flags; seal shell access; *set and freeze a calibrated alert profile*; pin a channel ceiling; run acceptance on the sealed image.

② A clear path to install updates SCOPE GREW

Where it stands — still a genuine strength: the project ships an offline update system with pre-checks, automatic backup, a verify step and a rehearsed one-command rollback. That machinery is real and tested.

What changed — materially: every release until now was a code overlay against a frozen schema, and the update tooling was built for exactly that. 0.8.5 requires three database migrations, forbids the old field-upgrade recipe, and ships an updater that refuses to migrate by design. On a box with no customer shell, the current answer to "how does a fielded appliance take 0.8.5?" is that there isn't one yet.

The July gaps remain on top of that: the offline bundle is still built for Intel/AMD rather than the Spark's arm64, bundles are not cryptographically signed, and there is no CI to build them reproducibly — so authenticity rests on process rather than proof.

Actions: choose the appliance migration model (updater gains migration rights · whole-volume A/B swap · manufacture-time only); build and *test a database rollback*, not just a code rollback; produce a signed arm64 artifact; stand up CI to build it reproducibly; keep the preflight → apply → verify → rollback chain as the on-box mechanism. **Re-estimate before committing to a date.**

③ Python code unreachable by anyone but the EVA team

Where it stands: unchanged since July, and the manual findings sharpened one edge of it — service credentials, including a video-system account holding administrator rights, sit in plain text in the environment file. As a source install the box offers no protection at all: acceptable for a lab, not for a field unit.

The layered approach still applies, and no single measure is sufficient: no customer shell or console access; full-disk encryption keyed to the box's hardware root of trust; the load-bearing modules compiled to native binaries so disk access yields machine code rather than source; and secrets sealed to the hardware rather than sitting in a readable file.

One thing got easier: the secret surface was enumerated and is small — four values, three of which are database connection strings. Sealing four values is a contained piece of work, not an audit.

Actions: choose the compilation toolchain and confirm it runs on arm64; add it to the image build; adopt disk encryption sealed to the box; move the four secrets into that sealed store; rotate the video-system credential, which has transited lab shell history. Substantial — still the largest new workstream.

④ Hardware locking — no cloning; only boxes from or authorised by us

Where it stands: unchanged. Nothing binds the software to a machine; a copied disk would run anywhere. The software's offline, no-phone-home design — a strength for air-gap — means the lock must also be offline, with no activation server.

The shape: at manufacture, issue a licence file signed by an EVA private key naming that specific box by hardware serial, network and disk identifiers, ideally sealed to its security chip. At every start the appliance verifies the signature and that the fingerprint matches the hardware it is running on, refusing to start on a mismatch. A cloned disk on a different box fails the check.

Coupling to note: the lock is only as strong as the code protection hiding it, so ③ and ④ stay one workstream. If the hardware target moves off the Spark, the fingerprint definition and the security-chip question both reopen — another reason to settle the platform question early.

⑤ Control load and performance; forewarn and suggest best settings GOT EASIER

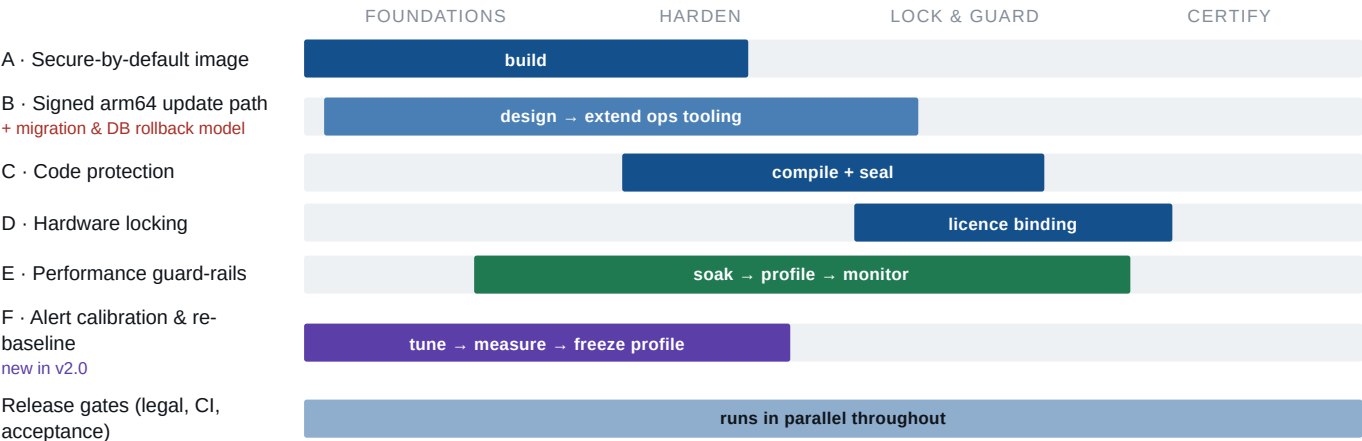
This is where the last three days paid the largest dividend. In July this point was well understood but entirely unmeasured. It now has a real baseline — roughly 110 summaries per channel-hour, evenly distributed across nine channels, with sub-second timing — and a repeatable two-hour method for re-measuring it.

And 0.8.5 supplies the missing telemetry. The readiness endpoint now exposes attention storage state, queue, drop and failure counters, coordinator status and active probe counts. The guard-rail monitor that July described as new engineering is now largely a matter of surfacing what the platform already reports.

The warning signal is already visible. Drop counters non-zero on every channel at nine channels, and a separate observation of vision throughput declining system-wide over hours (one channel from ~109 to ~69 per hour). Whatever the mechanism, that is exactly the trend an early-warning surface exists to catch — and it argues for the monitor being built before the soak rather than after it.

Actions: soak at rising channel counts to set the published ceiling; encode a Spark performance profile as shipped defaults; build the monitor on 0.8.5's counters; add the throughput-over-time trend to it, since a slow decline is invisible in any instantaneous reading.

INDICATIVE SEQUENCING TOWARD RELEASE 1.0 — RELATIVE EFFORT AND ORDERING, NOT CALENDAR-COMMITTED



C and D remain coupled — hardware locking is only as strong as the code protection that hides the check. E can begin as soon as one box runs a realistic channel load. **F is new in v2.0 and sits early on purpose:** the calibrated alert profile has to be frozen before the image that ships it is defined, and re-baselining after 0.8.5's new batching contract is a prerequisite for any capacity claim.

Net assessment. The strategy is sound and the software is a good fit for it. Point ⑤ moved from the hardest-to-evidence point to the best-evidenced one. Points ③ and ④ are unchanged and remain the real new build. Point ② is the one that grew: it is no longer an extension of existing tooling but a design problem about migrating a database inside a sealed box. And one workstream was added — calibration — because a sealed appliance ships its defaults forever.

What needs a decision, and what happens next

DECISIONS ONLY STAKEHOLDERS CAN MAKE

#	QUESTION	BLOCKS RELEASE?
1	Which software licence will the product ship under, and is the one unknown-licence dependency acceptable or replaced?	Yes
2	Do the AI model licences permit commercial government surveillance use? Written confirmation needed.	Yes
3	Is the current branch — thirteen commits past the last reviewed point — the intended release line, and are those commits reviewed? It is also the box 0.8.5 must migrate <i>from</i> .	Yes
4 NEW	Is the Spark still the appliance target? The 0.8.5 notes name a 4070-class host and a "port deployment" as the capacity reference. Points ①, ③ and ④ all assume the platform.	Yes — scopes ① ③ ④
5 NEW	What is the appliance migration model? Updater gains migration rights, whole-volume A/B swap with rollback, or migrations only at manufacture. Determines the size of point ②.	Yes — scopes ②
6 NEW	Who owns alert policy? Severity mapping, detection thresholds and probe floors are product decisions with no named owner. A sealed appliance ships whatever is decided, permanently.	Scopes work
7	Is EVA AI multi-tenant in practice? If not, several security findings downgrade; if so, the shared skills folder needs scoping.	Scopes work
8	Code-protection depth: compiled binaries, or sealed-image-plus-no-access as sufficient? Sets the size of the largest workstream.	Scopes work
9	Does the target box expose a hardware security chip we can seal keys to? Determines how tamper-resistant locking can be.	Informs ④

RECOMMENDED IMMEDIATE NEXT STEPS

- Finish Phase-6 acceptance on the Spark.** It is the open gate from the install, the sign-off for the first image, and now the precondition for the 0.8.5 migration. It has been the top item for two weeks; everything else queues behind it.
- Get Sasha's read on the dropped batches.** Are 47 dropped batches in two hours at nine channels benign coalescing or silent evidence loss? One answer either closes the top "should fix" item or makes the durable queue mandatory before any release.
- Close the four legal and honesty blockers.** Add a licence and notice file, obtain model-use confirmation, flip secure defaults, correct the audit-log wording. Cheap, unchanged since July, and they unblock everything else.
- Settle the platform and migration-model questions** (decisions 4 and 5) before any image work starts. Both change what gets built rather than how fast.
- Calibrate the alert profile and re-baseline on 0.8.5.** The bench re-measures in about two hours; 0.8.5's new batching contract invalidates the current numbers anyway, so the re-baseline is required regardless.
- Stand up CI, now including an upgrade-and-rollback test.** The suite and checks already exist; wiring them to run automatically is still the highest-leverage continuity fix, and migrations make the upgrade test newly necessary.

Confirmed at this milestone

The Spark install is live and green — the first arm64 deployment. The software makes no external calls and carries no telemetry. The agent cannot change state without a human, and 0.8.5 bounds its work by construction. Tenant data is isolated in the database. The minimal camera-system permissions the product actually needs are known. And as of this week the detection chain has been measured end to end against ground truth with sub-second timing accuracy and zero false positives. These are real foundations to build the appliance on.

Sources. Seven-part read-only review of the EVA AI source at commit [beca6ff](#) (24 July 2026); Wave-1 end-to-end benchmark, run [2026-07-25-e2e8](#), nine channels over 2.07 h at default settings, with the Feature Pack 1 and Part A experiments; manual QA findings MF-1...MF-12 (24–25 July); Luxriot EVA AI β 0.8.5 release notes (27 July 2026). Full technical reports and the companion findings register are retained alongside this document; every rating and figure is traceable to them.